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AI-Powered Smart Research Repository

Information System
CIFOR-ICRAF

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WHY

AI-Powered Smart Research Repository

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W H Y ? AI-Powered Smart Research Repository

Information Overload

Too many research journal documents to read one by one

Keyword Limitation

Conventional search in SQL Server only matches exact keywords, rather than searching by intent or context.

Time Consumption

Researchers spend too much time merely searching for relevant references

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SO,

What's the solution?

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The “SMART” Concept

Automated Tagging

AI identifies topics and metadata without manual input

Semantic Search

Search by semantic meaning or concepts, not merely keyword matching

Knowledge Graph

Mapping relationships between research that were previously unseen.

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HOW,

Bringing the Vision to Life

All research document sources such as publications, reports, papers, and internal archives are automatically collected and processed as AI model learning data.

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Intelligent Data Ingestion

All research document sources such as publications, reports, papers, and internal archives are automatically collected and processed as AI model learning data.

The AI Processing Core

The core of AI processing with the Retrieval-Augmented Generation (RAG) approach.

Seamless Insight Delivery

The final result provided by AI is in the form of a summary, labeled as a list of document files.

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Intelligent Data Ingestion

Text extraction process

Text Extract

Text Processing

Text cleaning, format correction, character normalization, and metadata noise removal

Chunking Engine

The process of chopping or cutting text into smaller pieces before converting it into vector embedding

Embedding Vector

This method converts text into a high-dimensional numerical representation that captures the semantic meaning of sentences and generates vector embeddings.

Vector Database

a specialized type of database designed to store, manage, and search data in the form of vector embeddings

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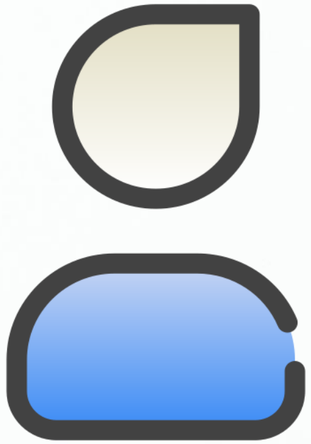
The AI Processing Core



Input Question

Query Embedding

The process of converting user questions into vector embeddings

Vector Similarity Search

Search for the most relevant context based on vector similarity. The vector from the query is compared with vectors in the vector database

Context Augmentation

Combining search results with prompts. Relevant documents found will be inserted into the prompt as additional context before being sent to the LLM.

LLM Generation

AI generates answers. After receiving questions and supporting information, AI compiles appropriate and relevant answers.

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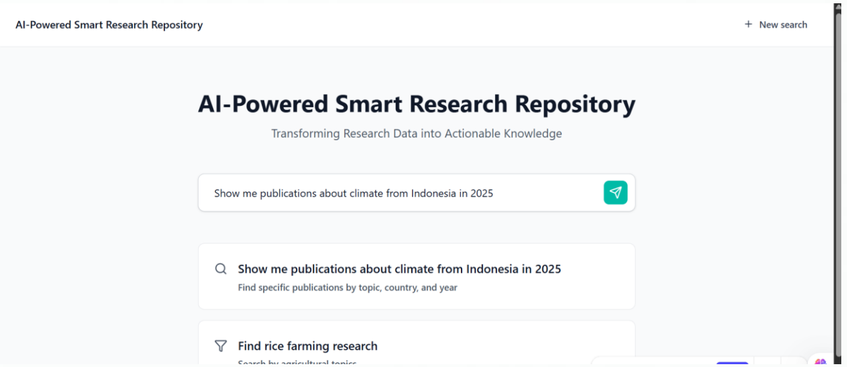
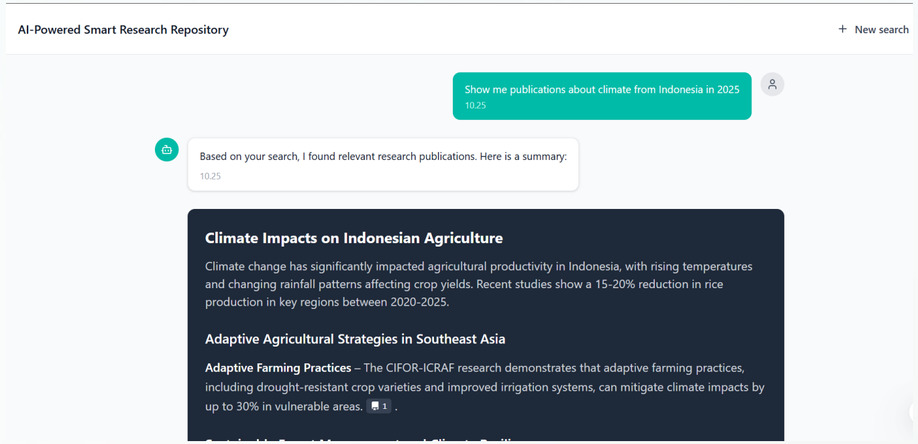
Seamless Insight Delivery





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THE IMPACT,

Return on Investment

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Core Capabilities

Summarization

Generate executive summaries from thousands of pages of documents instantly.

Cross-Language Synthesis

The ability to search and summarize literature in multiple languages.

Contextual Q&A

A chatbot feature that allows researchers to 'dialogue' with their research database.

Trend Analysis

Detect emerging research patterns in the future.

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Value & Efficiency

Enhanced Collaboration

Cross-division teams can access the same understanding without duplicating work.

Evidence-Based Decisions

More accurate decision-making because it is supported by easily accessible data

Accelerated Research

Accelerate the literature review process

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Our goal is not to replace researchers with AI, but to free them from the administrative task of searching for data, so they can focus on what matters most: conducting better research, faster.